ARPIT K. JAISWAL

GAMEPLAY & AI SYSTEMS DEVELOPER [UNITY DEVELOPER] [3.5+ YRS EXPERIENCE]

+91 8149135849 • arpitjaiswal071@gmail.com Portfolio ↗ | LinkedIn ↗ | GitHub ↗ | GitLab ↗

EXECUTIVE PROFILE

Core Specializations & Studio Focus

- **Production & Multiplayer Systems (Unity C#):** Gameplay & Systems Engineer with 3.5+ years shipping 3 commercial Google Play titles and real-time multiplayer games using Photon (PUN 2), Stockfish AI, and deterministic FSMs.
- **Architecture, Native Bridges & 60 FPS Optimization:** Specialist in procedural N×N mesh/puzzle generation, native Android (JNI) / iOS (Obj-C) bridges, non-blocking OTA content delivery, and Zero-GC object pooling locked at 60 FPS.
- **AI-Augmented Tooling & LiveOps Velocity:** Leverages Claude, ChatGPT/Codex, and custom Unity EditorWindows to accelerate URP/HLSL shader prototyping, offline localization pipelines, and Google/Apple IAP monetization.

PRODUCTION TRACK RECORD (KPIS)

Quick 6-Second Recruiter Highlights

3.5+ Years — STUDIO EXPERIENCE

- **Vasundhara Game Studio, Surat (2yr)** • **Dec 2024 – Jan 2025 & Dec 2025 – Present**
- **StudioKrew, Delhi (3m)** • Mar 2024 – May 2024
- **Appzia, Pune (1yr)** • June 2024 – July 2025

3 Shipped Titles — GOOGLE PLAY

- **Calm Jigsaw** [Store ↗]
- **MeowSweeper** [Store ↗]
- **Calm Solitaire** [Store ↗]

Engineered Subsystems [Unity Playable Demo ↗]

- **Procedural Jigsaw**
- **Jigsaw Shader Mask**
- **Native Image Bridge**
- **Level Sync (OTA)**
- **Live-Ops Backend**
- **Offline Leaderboard**
- **IAP Currency State**
- **Argos Localization**
- **Asset Organizer Tool**
- **Arrow Strike**

Early-Career Prototypes [GitLab Repo ↗] [Demos ↗]

This Is Blast • Arrow Strike • Tank Battle • Bullet Chess • Tycoon RTS • Trivologist

PROFESSIONAL STUDIO EXPERIENCE

Commercial Production Roles & Systems

Vasundhara Game Studio | AI Product Developer — Unity Systems & Gameplay [CURRENT]

Surat, GJ • **Dec 2024 – Jan 2025 & Dec 2025 – Present**

- **Procedural Jigsaw & Shader Mask (Calm Jigsaw):** Architected N×N Bezier puzzle generator using 9 mathematical edge signatures, dynamic UV quad mesh generation, and custom URP/HLSL shader masking with zero pre-sliced art.
- **Native Image Bridge & OTA Sync:** Engineered C# bridge over native Android (JNI) / iOS (Obj-C) APIs for 3-tier RAM LRU and disk caching, texture downsampling, and non-blocking OTA manifest level delivery.
- **Hybrid Logic & Hint-Cascade Solver (MeowSweeper):** Built hybrid Sudoku/Minesweeper/Star-Battle puzzle engine with backtracking hint-cascade solver FSM, discrete multi-touch locks, and persistent IAP powerup wallet.
- **Card Seam Merging & Arrow Mechanics (Calm Solitaire & Arrow):** Engineered rectangular card-grid 0px seam merging, displacement swapping, procedural flame noise shaders, and outside-in peeling arrow escape mechanics.
- **Offline Leaderboard, IAP & Editor Tooling:** Built zero-server bot-paced seasonal leaderboards with anti-rollback clamping, Google/Apple IAP state machines, Asset Organizer EditorWindow, and Argos IPC localization.

PROFESSIONAL STUDIO EXPERIENCE (CONTINUED)

Multiplayer, AI & Standalone Game Architecture

Appzia | Unity3D Systems & Gameplay Developer

Pune, India • June 2024 – July 2025

- **Stockfish AI & Real-Time Chess (Bullet Chess):** Architected multiplayer chess engine using Photon (PUN 2) and Stockfish API for live move evaluation, client-side board validation, clock sync, and tournament brackets.
- **Arcade Simulation & Economy Systems (Pinnacle):** Led development of Pinnacle arcade simulation replicating physical machines, implementing deterministic FSMs, secure Apple/Google IAP, and Zero-GC object pooling at 60 FPS.

StudioKrew | Unity Game Developer (Remote)

Delhi, India (Remote) • March 2024 – May 2024

- **Solo Card Game Architecture (FreeCell & Solitaire):** Delivered complete end-to-end development independently, architecting deterministic card stack rules, memento-based undo/redo state history, and responsive touch controls.
- **Dice Physics & Scoring State Machine (Yahtzee):** Engineered probabilistic dice-rolling physics, category score validation state machines, and turn-based round flow with fluid particle animations.
- **Grid Puzzle Engine & Serialization (BrokenTiles):** Developed reactive grid-matching algorithms with cascading tile clearing, implementing lightweight JSON save serialization for instant session recovery.

TECHNICAL SKILLS & ENGINE STACK

Production Engineering & Tooling Matrix

Core Engine & Languages

Unity 3D (2020–2023 LTS / Unity 6) • C# (.NET) • HLSL / ShaderLab • Python

Rendering & Optimization

URP • SRP Batcher • GPU Instancing • ASTC Compression • Custom UI Shaders

Profiling & Debugging

Unity Profiler • Memory Profiler • Frame Debugger • RenderDoc • Android Logcat / ADB

Platform & Native Bridges

Android (JNI / Java) • iOS (Obj-C Bridge) • WebGL (WASM) • OTA Manifest Sync

Architecture & AI Systems

Deterministic FSMs • MVP / ScriptableObjects • Zero-GC Object Pooling • Stockfish / A* / Backtracking

LiveOps, SDKs & Tooling

Google / Apple IAP • Ads Mediation • Firebase • Custom EditorWindows • Git / GitLab

EDUCATION

Academic Qualifications & Computer Science Foundation

Master of Computer Application (MCA) [CGPA: 7.52]

Government Autonomous Engineering College

Aurangabad, Maharashtra **Dec 2021 – Jan 2023**

Bachelor of Computer Science (B.Sc. CS) [CGPA: 7.50]

Vivekanand Arts and Science College

Aurangabad, Maharashtra **Apr 2018 – Oct 2021**